MIL-P-51300
12 March 1969
(See section 6)

MILITARY SPECIFICATION

PHOSPHORUS TRICHLORIDE, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers analyzed reagent grade phosphorus trichloride.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-W-572 - Wood-Preservative; Water-Repellent.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-C-186 - Containers, Packaging and Packing For Drugs, Chemicals, and Pharmaceuticals.
PPP-C-300 - Chemicals, Liquid; Packaging and Packing of.

MILITARY

MIL-P-15011 - Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners or Pallet-Type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publications should cite "49 CFR 171-179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D86-62 - Distillation of Petroleum Products.
D891-59 - Specific Gravity of Industrial Aromatic Hydrocarbons and Related Materials.
D1193-66 - Reagent Water.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Physical and chemical characteristics. Phosphorus trichloride shall conform to the physical and chemical characteristics of table I when tested as specified therein.

Table I. Physical and chemical characteristics

Characteristic	Minimum	Maximum	Test paragraph
Specific gravity	1.54	1.60	4.2.4.1
Boiling range, °C	75.0	76.0	4.2.4.2
Sulfate (SO_4), percent by weight	-----	To pass test	4.2.4.3
Heavy metals (as Pb), percent by weight	-----	0.001	4.2.4.4
Iron (Fe), percent by weight	-----	0.0015	4.2.4.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the phosphorus trichloride produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 100 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken. Specimen bottles should be equipped with a self-sealing diaphragm to preclude entrance of air and should be purged with nitrogen. Transfer of phosphorus trichloride to the specimen bottle should be accomplished under a blanket of nitrogen. The use of nitrogen purged hypodermic syringes is recommended.

Table II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2-25	2
26-150	3
151-1,200	5
1,201-7,000	8
7,001-20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified.
(b) Container not as specified.
(c) Container closure not as specified.
(d) Container damaged or leaking.
(e) Marking incorrect, missing, or illegible.
(f) Palletization incorrect (where required).
(g) Wood preservative missing (level A only).

4.2.3.2 For test. Approximately equal portions of phosphorus trichloride shall be removed from the containers taken in 4.2.2.2 and shall be thoroughly mixed to form a composited specimen of no less than 400 g (Where less than five containers are sampled, the composited specimen shall weigh no less than 200 g). The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. (A single analysis shall be run when the composited specimen weighs 200 g). Failure of either analysis of any test shall be cause for rejection of the lot represented. (See specimen handling and transfer recommendations in 4.2.2.2).

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4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Transfer of phosphorus trichloride shall be accomplished under a nitrogen blanket. All test apparatus should be purged with nitrogen, and a nitrogen atmosphere maintained where applicable. Where applicable, blank determination shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Specific gravity. Determine the specific gravity in accordance with ASTM D891.

4.2.4.2 Boiling range. Determine the temperatures between which 95 milliliters (ml) of a 100-ml specimen distills in accordance with ASTM D86 and correct to 760 millimeters of mercury.

4.2.4.3 Sulfate. Transfer 2 ml of the specimen to a 150-ml beaker containing 75 ml of water. (Care must be taken when mixing phosphorus trichloride with water because of the heat produced.) Add concentrated ammonium hydroxide solution dropwise while stirring, making the specimen solution just alkaline to congo red paper. Add 2 ml of concentrated hydrochloric acid and heat to boiling. Add 5 ml of 10-percent barium chloride solution. No turbidity shall be produced in 5 minutes.

4.2.4.4 Heavy metals.

(a) Solution A. Dissolve 3 ml of the specimen in 30 ml of water. Add 2 ml of concentrated nitric acid and heat to boiling. Boil gently for no less than 15 minutes. Cool the solution and dilute to 50 ml with water.

(b) Procedure. Prepare a control by adding 0.02 milligrams (mg) of lead ion, 7 ml of 10-percent hydrochloric acid, and 4.0 g of anhydrous sodium acetate to 5 ml of solution A. Dilute to 40 ml with water. Add 4.0 g of anhydrous sodium acetate to 25 ml of solution A and dilute to 40 ml with water. To each solution, add 10 ml of saturated hydrogen sulfide water. Mix thoroughly. Any brown color produced in the specimen solution shall be no darker than that produced in the control.

4.2.4.5 Iron.

(a) Butanolic potassium thiocyanate solution. Dissolve 10 g of potassium thiocyanate in 10 ml of water. Warm the solution to 27° + 3° C. Dilute with n-butyl alcohol to 100 ml and shake vigorously until clear.

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(b) Procedure. Transfer 17 ml of solution A, prepared as specified in 4.2.4.4 (a), to a 60-ml separatory funnel. Add 1 ml of concentrated hydrochloric acid and 50 mg of ammonium persulfate. In another 60-ml separatory funnel, prepare a control by adding 2 ml of concentrated hydrochloric acid and 50 mg of ammonium persulfate to 0.025 mg of iron ion and diluting to 18 ml with water. To each funnel, add 15 ml of butanolic potassium thiocyanate prepared as specified in (a) above. Shake each funnel vigorously for no less than 30 seconds. Allow the layers to separate. Any red color produced in the butanol (upper) layer shall be no darker than that produced in the control.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One-quarter pound ± 0.05 ounce or 1 pound ± 0.1 ounce as specified (see 6.2) of phosphorus trichloride shall be packaged in a bottle conforming to type I class 1, of PPP-C-300, with closure H of PPP-C-186, and utilizing inclosure b of PPP-C-300.

5.1.1.1 Intermediate packaging. Twelve 1/4-pound bottles or 12 1-pound bottles of phosphorus trichloride, packaged as specified in 5.1.1, shall be intermediately packaged as specified in PPP-C-300, except that only one three-bottle by four-bottle tier shall be placed in each intermediate container.

5.1.2 Level C. Specified quantities of phosphorus trichloride shall be packaged in type I, class 1 bottles with inclosure A of PPP-C-300.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Four intermediate packages of 1/4-pound bottles or two intermediate packages of 1-pound bottles, packaged as specified in 5.1.1.1, shall be packed upright in a nailed wood box, a cleated-plywood box, or a wirebound wood box, as specified (see 6.2). The nailed wood box shall conform to class 2, style 4-1/2, for a type 2 load of PPP-B-621. The cleated-plywood box shall conform to overseas type, style I or J, for a type 2 load of PPP-B-601. The wirebound wood box shall conform to class 3, style optional, for a type 2 load of PPP-B-585. Fiberboard fillers and pads shall be used as directed in PPP-C-300 to assure a tight pack. Each container shall be closed and strapped with galvanized strapping as specified in the appendix of the applicable box specification. Unless specific provision for preservation is made in the appropriate box specification, each box or finished, unpainted wood parts thereof shall be completely immersed for no less than 3 minutes in a water-repellent wood-preservative conforming to composition A of TT-W-572.

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5.2.2 Level B. Phosphorus trichloride shall be packed as specified in 5.2.1, except that a fiberboard box conforming to style RSC, grade W5c of PPP-B-636 shall be used in place of the wood box. No preservative shall be used. The box shall be closed and reinforced by pressure sensitive adhesive filament tape banding as specified in the appendix to PPP-B-636.

5.2.3 Level C. Twelve bottles of phosphorus trichloride, packaged as specified in 5.1.2, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Department of Transportation (DOT) Specifications 15A, 15B, 15C, 16A, or 19A.

5.3 Marking. In addition to the special marking specified herein, and to any special marking specified in the contract or order (see 6.2), unit, intermediate, and shipping containers shall be marked in accordance with DOT regulations and MIL-STD-129. The "Corrosive Liquid" white label specified by DOT regulations shall be applied to each intermediate and shipping container.

5.3.1 Special marking. Each unit container shall be durably and legibly labeled with the following information:

(a) Empirical formula
(b) Formula weight
(c) Manufacturer's lot number
(d) Assay
(e) Analysis or maximum limits of impurities
(f) Net weight of contents.
(g) Precautionary marking:

DANGER!

Causes severe burns.
Vapor extremely irritating.
Contact with water may cause flash fire.
Do not get in eyes, on skin, or on clothing.
Do not breathe vapor.
In case of contact, immediately remove all contaminated clothing and flush skin or eyes with plenty of water for at least 15 minutes; for eyes, get medical attention. Wash clothing before re-use.
In case of spillage, flood carefully with large volumes of water and provide adequate ventilation.
Do not add water to contents while in a container because of violent reaction and possible flash fire.
Do not open unless you have been properly instructed.
Do not clean or re-use this container.

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5.4 Palletization. When specified (see 6.2), shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147. A pallet conforming to style 1, type optional, class 1 of MIL-P-15011 shall be used. Galvanized steel strapping no less than 1.25 inch wide and 0.035 inch thick shall be used. A horizontal strap shall be applied to each tier. Tie-down straps shall be parallel to the 40-inch dimension of the pallet. The pallet and any other wood storage aids shall be treated with a wood preservative as specified in MIL-P-15011.

6. NOTES

6.1 Intended use. Phosphorus trichloride is intended for use as a laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and the date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Unit quantity required (see 5.1.1 and 5.1.2).
(d) Kind of box required for level A packing (see 5.2.1).
(e) Additional marking, if required (see 5.3).
(f) Palletization, if required (see 5.4).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.5 Safety precautions. Phosphorus trichloride is extremely hazardous on contact and will cause serious burns to eyes and skin. The vapors are extremely irritating. It decomposes rapidly on contact with atmospheric moisture. Contact with water may cause a flash fire. For additional information, refer to Manufacturing Chemists' Association Chemical Safety Data Sheet SD-27.

6.6 Supersession data. This specification includes the requirements of Edgewood Arsenal Purchase Description 4-192, dated 25 October 1952.

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Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD, MU(FA)
Air Force - 68
DSA - GS

User activities:

Army - MI, SM

Preparing activity:

Army - MU(EA)

Project No. 6810-0453

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA-QAES
Edgewood Arsenal, Maryland 21010

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